1️⃣ Identify whether a mathematical conclusion makes sense in context (Most Relevant)

Relevant item:

2. Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem interpret

How it is reflected

Kevin's reasoning is:

TV ownership percentages are similar
→ Norway has more households subscribing to cable TV.

Students must evaluate whether this conclusion is reasonable.

Looking at the data:

- Households owning TVs in **France**: **24.5 million**
- Households owning TVs in **Norway**: **2.0 million**

Although:

- France cable rate = **15.4%**
- Norway cable rate = **42.7%**

the total number of TV-owning households in France is far larger.

Calculating the number of cable subscribers:

France:

$$24.5 \times 0.154 \approx 3.77 \text{ million}$$

Norway:

$$2.0 \times 0.427 \approx 0.85 \text{ million}$$

Therefore:

France actually has more households subscribing to cable TV.

Students must recognize that Kevin's conclusion **does not make sense given the data**.

2️⃣ Interpret mathematical outcomes in relation to the situation

Relevant item:

5. Interpret mathematical outcomes in a variety of formats in relation to a situation interpret

How it is reflected

The information is presented in a **table format**, including:

- Quantities (millions of households)
- Percentages

Students must understand that the third column represents:

the percentage of TV-owning households that subscribe to cable TV,
not the percentage of all households.

They must also interpret how **quantities and percentages work together** to represent real-world information.

3️⃣ Use contextual knowledge to judge the reasonableness of results

Relevant item:

6. Use knowledge of how the real world impacts the outcomes and calculations of a mathematical procedure interpret

How it is reflected

Students must understand a real-world principle:

A higher percentage does not necessarily mean a larger total.

Although Norway has a higher subscription rate:

- Norway = **42.7%**

its base population is much smaller:

- only **2 million households**

France has a much larger base:

- **24.5 million households**

Thus:

A higher percentage does not imply a larger total number of subscribers.

4️⃣ Understand the relationship between context and mathematical representation

Relevant item:

9. Understand the relationship between the context of the problem and representation of the mathematical solution interpret

How it is reflected

Students must understand the relationship:

Number of cable subscribers
= Number of TV-owning households × Subscription percentage.

Mathematically:

France:

$$24.5 \times 0.154$$

Norway:

$$2.0 \times 0.427$$

This demonstrates the ability to connect **context → mathematical relationship → interpretation of results**.

5️⃣ Construct and communicate explanations in context

Relevant item:

7. Construct and communicate explanations and arguments in the context of the problem

interpret

How it is reflected

Students must explain the reasoning clearly.

For example:

Kevin's statement is incorrect because although Norway has a higher cable subscription percentage, France has far more households that own TVs. As a result, the total number of cable subscribers in France is actually larger.

This demonstrates the ability to **construct explanations and arguments within the context of the problem**.

1️⃣ Identify whether a mathematical conclusion makes sense in context（最相关）

对应条目：

2. Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem interpret

体现方式

Kevin 的推理是：

电视拥有比例相似
→ 挪威订阅有线电视的家庭更多

学生需要判断这一结论是否合理。

通过观察数据可以发现：

- 法国家庭拥有电视数量：**24.5 million**
- 挪威：**2.0 million**

虽然：

- France cable rate = **15.4%**
- Norway cable rate = **42.7%**

但由于法国拥有电视的家庭数量远远大于挪威，因此：

法国订阅有线电视的家庭数可能更多。

计算：

France:

$$24.5 \times 0.154 \approx 3.77 \text{ million}$$

Norway:

$$2.0 \times 0.427 \approx 0.85 \text{ million}$$

因此：

法国订阅有线电视的家庭更多。

学生需要判断 Kevin 的结论 **不符合实际数据。**

2️⃣ Interpret mathematical outcomes in relation to the situation

对应条目：

5. Interpret mathematical outcomes in a variety of formats in relation to a situation interpret

体现方式

题目数据以 **表格形式**呈现：

- 数量（million households）
- 百分比（percentage）

学生需要理解：

第三列表示：

订阅有线电视家庭占拥有电视家庭的比例

而不是：

占全部家庭比例。

同时需要将：

- 数量
- 百分比

结合起来解释真实含义。

3️⃣ Use contextual knowledge to judge the reasonableness of results

对应条目：

6. Use knowledge of how the real world impacts the outcomes and calculations of a mathematical procedure interpret

体现方式

学生需要理解现实逻辑：

虽然挪威的订阅比例更高：

- Norway = **42.7%**

但其基数非常小：

- 仅 **2 million households**

而法国：

- **24.5 million households**

因此：

比例高并不意味着总数更大。

这是典型的：

比例 vs 绝对数量的现实判断问题。

4️⃣ Understand the relationship between context and mathematical representation

对应条目：

9. Understand the relationship between the context of the problem and representation of the mathematical solution interpret

体现方式

学生需要理解：

现实关系：

订阅家庭数 =
拥有电视家庭数 × 订阅比例

数学表达：

France:

$$24.5 \times 0.154$$

Norway:

$$2.0 \times 0.427$$

这体现了：

情境 → 数学关系 → 解释结果。

5️⃣ Construct and communicate explanations in context

对应条目：

7. Construct and communicate explanations and arguments in the context of the problem interpret

体现方式

学生需要给出解释，例如：

Kevin 的结论是错误的，因为：

虽然挪威的订阅比例更高，但法国拥有电视的家庭数量远远多于挪威，因此法国订阅有线电视的家庭总数实际上更多。

这体现了 **在情境中构建解释与论证的能力。**